

Working Title

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Abstract

1 Introduction

LLM-as-a-judge has become a widely used paradigm for evaluating open-ended generations; to support criterion-grounded judgments and make final scores more interpretable and auditable, many methods elicit rationales before scoring (Liu et al., 2023; Kim et al., 2024). In scientific-writing evaluation, this rationale-first design is particularly relevant because judgments often depend on domain knowledge, textual evidence and multiple quality dimensions (Sadallah et al., 2025; Şahinuç et al., 2026; Zhang et al., 2026).

Existing evaluation methods have introduced rationales through prompting, supervised distillation and reinforcement learning, and have generally reported higher human correlation, judge agreement or generalization ability (Liu et al., 2023; Kim et al., 2024; Şahinuç et al., 2026). These results suggest that explicit rationales can help models organize textual evidence and scoring decisions under complex evaluation criteria. However, existing experiments often change rationale generation together with prompt format, training objective, optimization strategy and inference protocol, making it difficult to determine whether the gains come from rationales themselves or from the full evaluation configuration. For example, the gains of G-Eval depend jointly on evaluation-step prompting and its scoring mechanism; the improvements of SciRM also combine domain-preference optimization, reasoning/self-verification training and rationale-first outputs, leaving the independent effects of rationale supervision and rationale-first inference unclear (Liu et al., 2023; Şahinuç et al., 2026).

To help models organize evidence and assign scores according to explicit criteria, some scientific-

writing evaluators have incorporated rationales into the scoring process (Sadallah et al., 2025; Şahinuç et al., 2026). However, rationales do not necessarily improve model performance: rationale augmentation can degrade task performance (Zhu et al., 2025), while CoT gains are concentrated mainly in mathematical or symbolic reasoning tasks and remain inconsistent elsewhere (Sprague et al., 2025). To determine whether performance changes arise from learning rationales during training or generating rationales before scoring at inference time, existing attribution studies have examined both settings, but primarily on general reasoning or NLU tasks (Wadhwa et al., 2024; Shi et al., 2025). Scientific-writing evaluation differs from these tasks: models must integrate domain knowledge and make ordinal judgments from textual evidence under multidimensional rubrics. Existing evidence therefore does not establish how rationale supervision and rationale-first inference individually affect scientific-writing evaluation, or how they interact when combined.

Our work. In this work, we systematically investigate how rationale supervision and rationale-first inference affect LLM-based scientific-writing evaluation. We establish a unified controlled framework that crosses two supervision targets with two inference protocols across seven scientific-writing evaluation tasks. We then analyze the role of rationales through aggregate performance, sample-level prediction changes, and sentence-level errors, and develop targeted training methods based on the identified problems.

Contributions and Findings Our contributions are threefold. (C1) We construct a 2×2 factorial design that crosses score-only supervision (SO) and rationale-and-score supervision (RS) with direct-score inference (DS) and rationale-first inference (RF). The design comprises 28 experimental conditions, each evaluated with three random seeds,

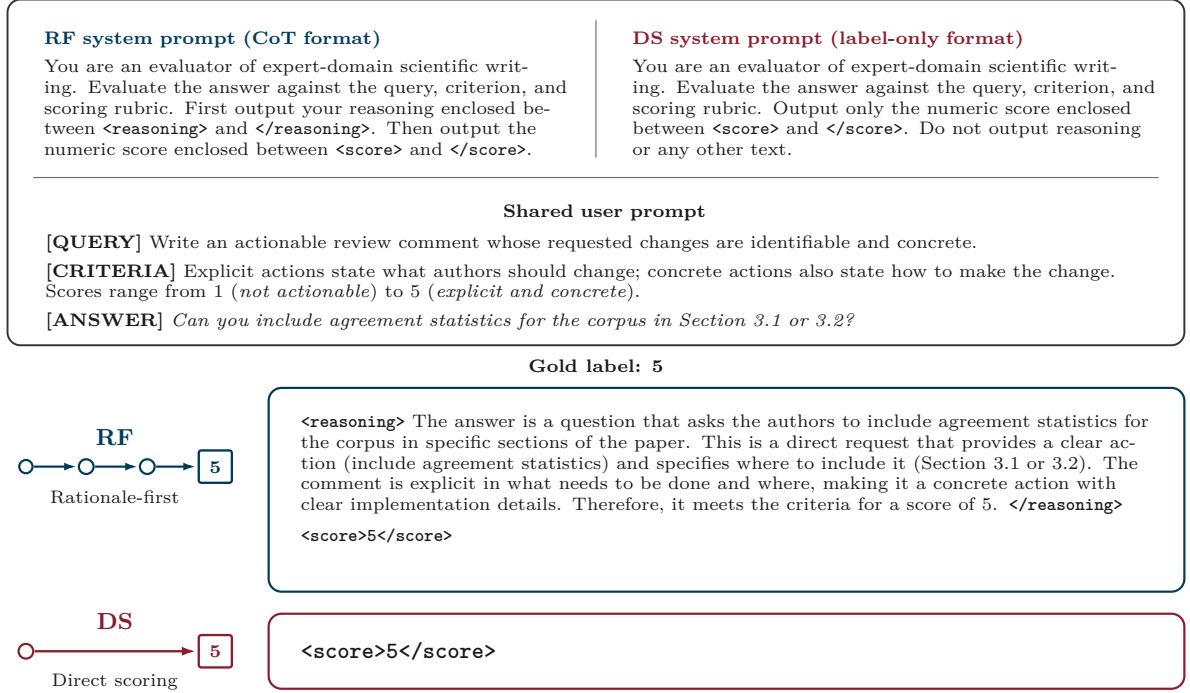


Figure 1: Matched direct-score (DS) and rationale-first (RF) evaluation prompts and outputs for the same scientific-writing example.

to separate the effects of supervision targets, inference protocols, and their interaction. **(C2)** We develop a multilevel analysis pipeline combining teacher-forced rationale NLL, rationale quality and consistency assessment, paired DS–RF prediction transitions, sentence-level error annotation, and controlled rescoring with fixed rationales. **(C3)** We introduce three targeted methods: Regenerated Rationales with Gold-Score Restoration (ReaSon) replaces teacher rationales with student-generated ones while restoring gold scores; Paired-Interface Reweighting (PaIR) trains on paired DS and RF views with equally weighted branch losses to address training–inference interface mismatch; and Regenerated-Rationale Paired-Interface Reweighting (RePaIR) combines the two designs.

Our experiments yield three main findings. **(F1)** RS improves teacher-rationale fitting, generated-rationale quality, and rationale–score consistency, but these gains do not translate into better scoring performance. On ordinal tasks, RF increases severe errors and lowers QWK under both supervision objectives, whereas its effects on binary tasks depend on the supervision objective. **(F2)** Rationales in harmful transitions typically support the incorrect score, with errors arising primarily from rubric misapplication and score mapping. Correcting er-

roneous rationales restores the gold score in 142 of 158 selected harmful cases, showing that rationale content can substantially affect subsequent scoring in these cases. **(F3)** ReaSon improves RF ordinal scoring but does not preserve DS performance; PaIR outperforms the matched Mix baseline under both inference protocols; and RePaIR preserves DS performance while increasing RF QWK from 0.698 to 0.735 and maintaining 100% strict-format validity under both interfaces.

2 Related Work

2.1 Evaluating Scientific Writing with LLMs

Scientific-writing evaluation is moving beyond generic text similarity toward task-specific quality judgments. Early work evaluates generated paper feedback through overlap with human reviews and author-perceived usefulness (Liang et al., 2024). For scientific text revision, LLM judges assess instruction following more reliably than correctness and therefore remain complementary to task-specific metrics (Jourdan et al., 2025).

Recent methods further assess different scientific-writing artifacts against explicit dimensions and scoring criteria. They cover review-comment utility, expert-preference-based related-work evaluation, citation-level scholarly

positioning, and multidimensional survey assessment (Sadallah et al., 2025; Şahinuç et al., 2025; Xie et al., 2026; Zhang et al., 2026). Cross-task evaluators also jointly generate rationales and scores under dynamically specified criteria and rubrics (Şahinuç et al., 2026). However, these methods typically change data construction, training objectives, and inference formats together. They therefore do not establish whether better rationale modeling improves scoring reliability or whether its effect depends on rationale-first inference.

2.2 Rationales in LLM Training and Inference

At inference time, LLM evaluators often generate a rationale before the final judgment. G-Eval elicits task-specific evaluation steps before scoring and reports stronger correlation with human judgments (Liu et al., 2023). However, later work shows that CoT can collapse the judgment distribution and reduce evaluation performance under common score readouts (Wang et al., 2025). Rationale-first inference therefore changes not only the reasoning context but also the path through which the score is produced and extracted.

At training time, teacher-generated rationales can serve as additional supervision. Prometheus jointly learns evaluative feedback and scores, whereas TRACT combines CoT supervision, regression-aware score learning, and self-generated rationales (Kim et al., 2024; Chiang et al., 2025). Although these methods report performance gains, they jointly change the training targets, loss functions, and rationale sources, leaving the independent contribution of rationale supervision unclear.

Controlled studies further show that the effect of rationales depends on the task, model scale, and output constraints. Consistent CoT gains are concentrated mainly in mathematical or symbolic reasoning (Sprague et al., 2025). On NLU tasks, inference gains vary with model scale, and most rationale-augmented training methods underperform label-only training (Shi et al., 2025). For fixed-format literary evaluation, reasoning supervision can also increase parsing failures and reduce overall performance (Liu et al., 2026). These studies, however, do not fully cross score-only and rationale-and-score supervision with DS and RF in ordinal scientific-writing evaluation, or jointly separate format failures from valid-score shifts.

3 Methods

We study model performance along three axes: supervision target, inference protocol, and rationale source. Training that supervises only the final score is referred to as score-only supervision (SO), whereas training that supervises both the rationale and final score is referred to as rationale-and-score supervision (RS). Direct-score inference (DS) outputs the score directly, while rationale-first inference (RF) generates a rationale before the score. Using standard RS as the baseline, we further compare three methods: ReaSon, PaIR, and RePaIR.

We first formalize standard RS. Let x_i denote the input of the i -th example, s_i its gold score, and r_i^T the rationale generated by the teacher model. The target completion is

$$y_i^R = \langle \text{reasoning} \rangle r_i^T \langle / \text{reasoning} \rangle \\ \langle \text{score} \rangle s_i \langle / \text{score} \rangle.$$

Standard RS averages token losses over the entire completion. Its objective can therefore be decomposed as

$$\mathcal{L}_{\text{RS}} = \frac{N_r}{N_r + N_s} \mathcal{L}_r + \frac{N_s}{N_r + N_s} \mathcal{L}_s,$$

where N_r and N_s are the numbers of rationale and score tokens, and $\mathcal{L}_r$ and $\mathcal{L}_s$ are their respective mean token-level losses. When $N_r \gg N_s$, the rationale span receives a larger weight in the overall scalar loss.

3.1 ReaSon: Regenerated Rationales with Gold-Score Restoration

ReaSon replaces teacher rationales with student-generated rationales while retaining the standard RS objective. Here, regeneration refers only to rationale regeneration followed by gold-score restoration. It does not involve an explicit critique-revision loop.

Teacher rationales r_i^T are distilled from deepseek-v4-pro and filtered for format validity and gold-score consistency, forming $\mathcal{D}_T = \{(x_i, r_i^T, s_i)\}$. A first-stage model θ_A is trained on $\mathcal{D}_T$ and then generates new rationales r_i^A under RF. Its predicted scores are retained only for recording, while the gold scores remain the training labels. This produces $\mathcal{D}_A = \{(x_i, r_i^A, s_i)\}$.

The second-stage model is reinitialized from the same base model rather than obtained by further updating θ_A . ReaSon therefore primarily replaces r_i^T with r_i^A as the source of rationale supervision.

The resulting model is evaluated under both DS and RF. Further details on data construction, prompts, and generation are provided in Appendix A.

3.2 Paired-Interface Reweighting (PaIR)

Standard RS has two limitations: long rationales receive greater weight under token-averaged training, and only the RF interface is observed during training. To address the former, we separately weighted the rationale and score regions within a single RF sequence and prevented the score region from attending to the rationale. However, this design still uses only RF prompts and therefore cannot resolve the DS interface mismatch.

PaIR constructs paired DS and RF training views for each example. The two views share the same user prompt and gold score but differ in their system prompts and output structures. The DS target y_i^D contains only the score block $\langle \text{score} \rangle s_i \langle / \text{score} \rangle$. The RF target y_i^R adds a rationale block $\langle \text{reasoning} \rangle r_i \langle / \text{reasoning} \rangle$ before the same score block.

The mean token-level losses of the two views are denoted by $\mathcal{L}_D$ and $\mathcal{L}_R$. They are computed separately and combined with equal weight:

$$\mathcal{L}_{\text{PaIR}} = 0.5\mathcal{L}_D + 0.5\mathcal{L}_R.$$

The resulting scalar loss is used for a single backward pass and parameter update. This objective balances the DS and RF views rather than the rationale and score tokens within the RF view.

We use Mix as a matched ablation. It shares the same data and training configuration as PaIR but removes branch-wise normalization. PaIR outperforms Mix under both DS and RF, indicating that branch-wise normalization provides additional gains.

3.3 Regenerated-Rationale Paired-Interface Reweighting (RePaIR)

RePaIR combines the rationale replacement of ReaSon with the paired supervision of PaIR. Starting from the base model θ_0 , the first stage trains a PaIR model θ_A on the teacher-rationale dataset $\mathcal{D}_T$. The resulting model then regenerates rationales r_i^A under RF. These rationales are paired with the gold scores s_i to form $\mathcal{D}_A = \{(x_i, r_i^A, s_i)\}$.

The second stage reinitializes the model from θ_0 and applies PaIR to $\mathcal{D}_A$, rather than further updating θ_A . The resulting model is evaluated under DS and RF, denoted by $\text{RePaIR}_{\text{DS}}$ and $\text{RePaIR}_{\text{RF}}$, respectively.

Comparing ReaSon with RS assesses the effect of rationale source. Comparing PaIR with Mix assesses branch-wise normalization. Comparing RePaIR with PaIR assesses the additional effect of replacing teacher rationales with student-generated rationales.

4 Experiments

Datasets. Our training data were adapted from (Şahinuç et al., 2026)’s work through task selection, prompt revision, and sample-level quality control. The resulting dataset covers two complementary dimensions of scientific-writing evaluation. Peer-review quality is scored on a five-point scale, while related-work assessment is formulated as binary classification. Further details on data processing and dataset statistics are provided in Appendix B.

Models. We used Qwen3-4B (Yang et al., 2025) as the base model for all experiments. For each task–method combination, we fine-tuned a separate model initialized from the same Qwen3-4B checkpoint.

Implementation details. We use LoRA to compare RS and SO supervision for scientific-writing evaluation. All seven datasets share the same training, validation, and test splits. Separate adapters are trained for each task and method, while the validation and test sets are used for model selection, analysis, and final evaluation. All evaluations use zero-shot prompts, with templates shown in Figure 1. LoRA is applied to all linear layers with $r = 16$, $\alpha = 32$, a learning rate of 1×10^{-4} , three epochs, cosine decay, and a maximum sequence length of 8,192 tokens. Additional settings are provided in Appendix A. Evaluating two supervision strategies across seven datasets and two inference protocols yields $2 \times 7 \times 2 = 28$ experimental conditions, each of which is run with three random seeds. All reported results are averaged across the three seeds (42, 43, and 44) and are presented in Section 5. We then analyze validation-set NLL and rationale quality under RF, followed by sentence-level analysis of prediction changes from DS to RF. For harmful changes, erroneous rationale content is replaced before rescoreing to assess its effect on the prediction. Finally, we evaluate ReaSon and PaIR separately before testing their combination, RePaIR.

4.1 Understanding the SO–RS Performance Gap

Teacher-forced Rationale NLL. To test whether the SO–RS performance gap is associated with teacher-rationale fitting, we compute teacher-forced rationale NLL for both methods. The analysis covers all tasks. Both methods use identical validation splits, comprising 1,368 samples and 161,178 teacher-rationale tokens in total.

For each adapter, complete prompt–rationale–score sequences are processed individually, while loss is computed only at rationale-token positions. Under causal masking, each token conditions only on the prompt and preceding teacher rationale tokens. We then micro-average the loss over all rationale tokens:

$$\text{NLL}_{\text{rat}} = \frac{\sum_n \sum_{t \in R_n} -\log p_{\theta}(r_{n,t} \mid x_n, r_{n,<t})}{\sum_n |R_n|}.$$

Here, R_n denotes the rationale-token positions in sample n . This yields three seed-level NLL values per method and task, which are averaged for reporting. A lower NLL indicates closer fitting to the teacher-rationale distribution.

Rationale Quality Comparison. Because NLL measures only teacher-rationale fitting, we further use external LLMs to assess generated-rationale quality and consistency with predicted scores. This analysis uses RF outputs from seed 42 and covers the same seven tasks. We randomly sample 50 paired SO–RS cases per task, yielding 350 pairs and 700 rationales.

GLM-5.3-Flash and Doubao-Seed-2.0-Lite serve as primary judges, with MiniMax-M3 as the adjudicator. Adjudication is triggered by invocation or parsing failure, judge disagreement, or a difference of at least two points on any dimension. It is used for 192 of the 350 pairs. The first round scores each rationale from 1 to 3 on rubric understanding, evidential support, key-information coverage, and unsupported-claim control. Mean scores determine whether one rationale is better or both are comparable. The second round classifies whether each rationale supports its predicted score as supported, partially_supported, or not_supported. Model configurations and evaluation prompts are provided in Appendix B.

4.2 Understanding the DS–RF Performance Gap on Ordinal Tasks

Across the 28 experimental conditions, the effect of the inference protocol varied by task type and training objective. DS consistently outperformed RF on ordinal tasks. On binary tasks, DS performed better under SO, whereas RF performed better under RS. We therefore compared sample-level DS-to-RF transitions, together with changes in Accuracy, QWK, and outcome distributions. The analysis covered four ordinal tasks and three random seeds, yielding 11,364 paired DS–RF predictions per training objective. Ordinal predictions were categorized as correct, adjacent errors, severe errors, or format-invalid.

For samples whose predictions changed, we applied the same three-judge protocol. Given the test instance, gold score, DS and RF predictions, and RF rationale, the judges determined whether the rationale supported the gold score, the incorrect prediction, or neither due to insufficient evidence. Harmful transitions were further annotated sentence by sentence for errors such as `factual_error` and `evidence_misread`. The complete taxonomy is provided in Appendix A.

Finally, we tested whether rationale correction changed subsequent scores in 158 harmful-transition cases. GLM-5.3-Flash corrected the original rationales, which were then reviewed by MiniMax-M3. Each case was rescored using its original adapter and the same RF system and user prompts. We denote natural RF generation as `NatRF`, rescored from the fixed original rationale as `FixOrig`, and rescored from the fixed corrected rationale as `FixCorr`. The `NatRF–FixOrig` comparison validates whether fixed-prefix rescored reproduces the original prediction, whereas `FixOrig–FixCorr` measures the score changes associated with rationale correction.

5 Results

All primary analyses use strict parsing to evaluate both scoring performance and format compliance, accepting only schema-conforming scores; relaxed-parsing results are reported in Appendix A.

RQ1: How Do SO and RS Differ in Scoring Performance and Rationale Modeling? Figure 2 (left) reports QWK on the four ordinal tasks. With score-only training, direct-score and rationale-first inference achieve mean QWK scores of 0.747 and

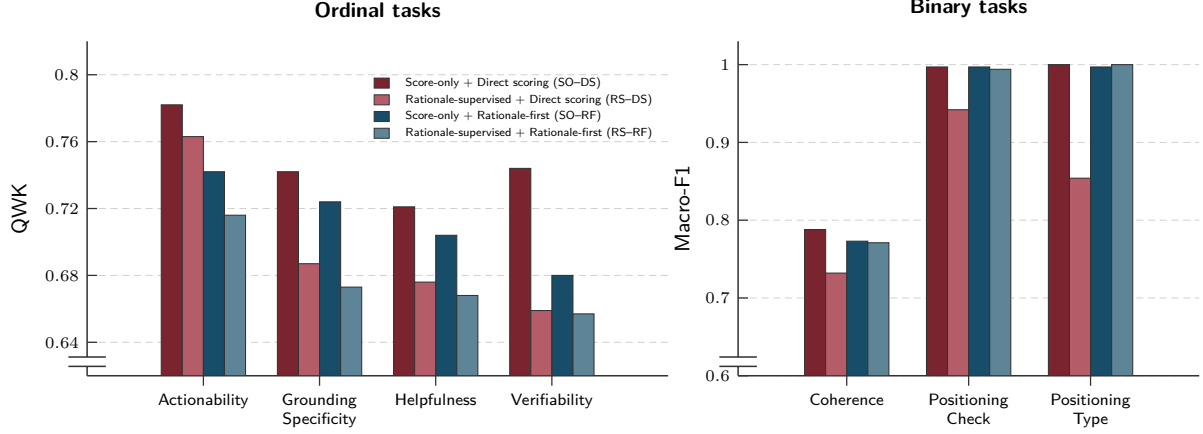


Figure 2: Performance across training objectives and inference protocols. The left panel reports QWK on ordinal tasks, and the right panel reports Macro-F1 on binary tasks.

0.712, respectively. With rationale-and-score training, the corresponding scores are 0.696 and 0.679. This pattern is consistent across tasks: score-only training outperforms rationale-and-score training, and direct-score inference outperforms rationale-first inference.

Figure 2 (right) reports Macro-F1 on the binary tasks. With score-only training, direct-score and rationale-first inference perform similarly, scoring 0.928 and 0.923, respectively. With rationale-and-score training, rationale-first inference reaches 0.922, compared with 0.843 under direct-score inference. The effect of the training objective therefore depends on the inference protocol, with a clear score-only advantage appearing only under direct-score inference.

Rationale supervision improves teacher-rationale fitting, but this advantage does not translate into better scoring. We measure teacher-rationale modeling using teacher-forced rationale NLL. Table 1 reports the task-level NLL results for Base, score-only training, and rationale-and-score training, where lower values indicate closer fitting to the teacher-rationale distribution. As shown in Table 1, the original Qwen3-4B without task-specific fine-tuning (Base) obtains a mean NLL of 3.4480, which decreases to 1.6964 with score-only training and 0.7793 with rationale-and-score training. Rationale-and-score training achieves the lowest NLL on every task, showing that direct rationale supervision improves teacher-token prediction but does not guarantee better scoring performance.

Rationale-and-score training produces higher-quality rationales. External LLMs compared 350 paired rationales. Table 2 reports the overall pref-

Task	Base	SO	RS	RS – SO
Actionability	3.3984	1.5891	0.7894	−0.7997
Grounding Specificity	3.6439	1.6757	0.6761	−0.9996
Helpfulness	3.5397	1.7811	0.8856	−0.8955
Verifiability	3.5677	1.9243	0.8401	−1.0842
Coherence	3.4272	1.4634	0.7852	−0.6782
Positioning Check	3.1925	1.5362	0.7112	−0.8250
Positioning Type	3.3664	1.9051	0.7676	−1.1375
Average	3.4480	1.6964	0.7793	−0.9171

Table 1: Teacher-forced rationale NLL by task; lower is better.

erence results, scores across four rationale-quality dimensions, and rationale–score consistency rates. As shown in Table 2, rationales from rationale-and-score training were preferred in 248 cases (70.9%). Score-only rationales were preferred in 25 cases (7.1%), while 77 cases (22.0%) were comparable. This preference was consistent across all evaluation tasks. Rationale-and-score training scored higher on rubric understanding, evidence grounding, key-information coverage, and unsupported-claim control, with the largest difference in key-information coverage. Its mean score across the four dimensions was 2.832, compared with 2.400 for score-only training, and its rationale–score consistency was also higher at 96.6% versus 74.3%.

RQ2: How Do DS and RF Differ in Ordinal Scoring? To characterize how the inference protocol affects ordinal scoring, we paired direct-score and rationale-first predictions for the same samples under both supervision objectives, yielding 11,364 prediction transitions. Predictions were categorized as correct, adjacent errors, severe errors, or format-

Analysis	Measure	SO	RS	Comparison
Pairwise Preference	Preferred rationale, n (%)	25 (7.1)	248 (70.9)	Tie: 77 (22.0)
Quality	Rubric understanding	2.379	2.848	+0.469
	Evidence grounding	2.495	2.837	+0.342
	Key-information coverage	2.273	2.833	+0.560
	Unsupported-claim control	2.453	2.811	+0.358
	Four-dimension mean	2.400	2.832	+0.433
Consistency	Supported (%)	74.3	96.6	+22.3 pp
	Partially supported (%)	20.0	3.1	−16.9 pp
	Not supported (%)	5.7	0.3	−5.4 pp

Table 2: External-LLM evaluation of 350 paired rationales generated under rationale-first inference. Quality dimensions use a 1–3 scale. For pairwise preference, Comparison reports tied judgments; elsewhere, it reports RS minus SO, with percentage differences in percentage points.

Training	Δ QWK	Δ Acc.	Δ Adj.	Δ Severe	Δ Invalid
SO	−0.035	−3.52 pp	−0.52 pp	+3.98 pp	+0.07 pp
RS	−0.017	+3.49 pp	−7.51 pp	+4.26 pp	−0.23 pp

Table 3: Macro-averaged changes from direct-score to rationale-first inference on the ordinal tasks. Prediction-category changes are reported in percentage points.

invalid. Table 3 summarizes the average changes after switching from direct-score to rationale-first inference.

With score-only training, QWK decreased by 0.035, accuracy by 3.52 percentage points, and the severe-error rate increased by 3.98 points. With rationale-and-score training, accuracy increased by 3.49 points, but adjacent errors decreased by 7.51 points while severe errors increased by 4.26 points, resulting in a QWK decrease of 0.017. Because QWK penalizes distant errors more heavily, the shift toward severe errors offset the gain in exact predictions. Invalid-output rates changed by less than 0.25 percentage points under both training objectives.

RF rationales closely align with prediction transitions, and their errors often emerge early. Across 11,364 DS-to-RF prediction pairs per objective, score-only training produced 377 harmful and 151 beneficial transitions, whereas rationale-and-score training produced 338 and 269. Judging was completed for 1,119 of 1,135 selected cases, with detailed counts reported in Appendix A.

Among 701 harmful transitions, 700 rationales supported an incorrect score. In contrast, all 418 beneficial-transition rationales supported the gold score. This result is consistent with the rationale–score alignment observed in RQ1.

Figure 3 summarizes the types and positions of

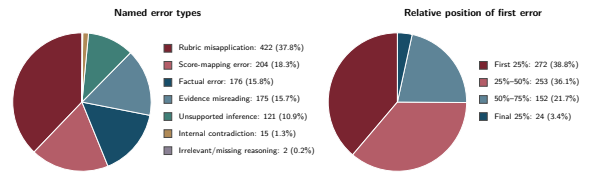


Figure 3: Sentence-level errors in harmful DS-to-RF transitions. The left panel shows the distribution of 1,115 sentences assigned a definitive error label; the right panel shows the relative position of the first error across 701 harmful transitions.

sentence-level errors. Across the two training conditions, rubric misapplication (37.8%) and score-mapping errors (18.3%) were the two most common types among 1,115 definitively labeled error sentences, together accounting for 56.1%, which indicates that errors primarily arise from applying the rubric and mapping the resulting judgment to a score. When the two training conditions are combined, 74.9% of first errors occur in the first half of the rationale, including 38.8% in the first quarter, suggesting that errors typically emerge early and may continue to shape the subsequent reasoning and final score.

Rationale correction substantially improved scoring in the selected harmful cases. FixOrig reproduced the NatRF prediction in 157 of 158 cases, yielding a 99.37% reproduction rate. Both settings produced valid scores, and their MAE values were nearly identical (2.038 versus 2.044), indicating that fixed-prefix rescoring rarely changed the original prediction. Full task-level reproduction results are provided in Appendix A.

Relative to FixOrig, FixCorr changed all 158 predictions. As shown in Table 4, 142 cases (89.87%) returned to the gold score, while the remaining 16

Task	RS		SO	
	<i>n</i>	FixCorr	<i>n</i>	FixCorr
Actionability	22	95.45	25	96.00
Grounding Specificity	25	96.00	18	100.00
Helpfulness	4	100.00	–	–
Verifiability	35	74.29	29	86.21
Overall	86	87.21	72	93.06

Table 4: Gold-score recovery after rationale correction in 158 selected harmful cases. Values report FixCorr accuracy (%); all FixOrig predictions are incorrect by construction.

improved from severe to adjacent errors. MAE decreased from 2.044 to 0.101. Among the 157 cases in which FixOrig reproduced NatRF, 141 (89.81%) recovered, showing that the aggregate improvement was not driven by the single reproduction failure. Recovery was observed across all evaluated task–training groups, showing that rationale correction consistently guided the model toward more accurate scores. The lower recovery rate for Verifiability indicates that the magnitude of improvement varies across tasks. Overall, erroneous rationale content was an important source of harmful predictions, and correcting it substantially improved subsequent scoring.

RQ3: How Can RS Preserve Rationale Quality and Scoring Performance? To address the rationale-source problem and the mismatch between training and inference interfaces identified above, we introduced ReaSon and PaIR. ReaSon modifies the source of rationale supervision, whereas PaIR improves adaptation to both DS and RF through paired-interface supervision. Because the two methods target complementary aspects of RS, we further combined them into RePaIR. Figure 4 shows the changes in ordinal QWK and binary Macro-F1 for all three methods relative to standard RS under the same inference protocol. Overall, RePaIR retained DS performance while achieving the largest improvement in RF ordinal scoring, providing the most balanced performance across the two inference protocols.

PaIR improves both interface coverage and branch learning through paired supervision. PaIR achieved ordinal QWK, MAE, and classification Macro-F1 scores of 0.757, 0.476, and 0.934 under DS, and 0.698, 0.555, and 0.929 under RF, with 100% strict-format validity at both interfaces.

Relative to Mix using the same paired data, PaIR increased QWK by 0.015 and 0.008, reduced MAE by 0.022 and 0.016, and increased Macro-F1 by 0.010 and 0.004 under DS and RF, respectively. These results show that paired supervision provides DS coverage, while branch-wise normalization offers an additional scoring benefit.

RePaIR combines the complementary gains of ReaSon and PaIR. Under DS, RePaIR achieved a QWK of 0.758, an MAE of 0.476, and a Macro-F1 of 0.931, closely matching PaIR at 0.757, 0.476, and 0.934. Under RF, RePaIR increased PaIR’s QWK from 0.698 to 0.735 and reduced MAE from 0.555 to 0.502, while Macro-F1 changed only from 0.929 to 0.926. Both interfaces retained 100% strict-format validity. RePaIR therefore preserves PaIR’s DS capability while concentrating the benefit of rationale regeneration on RF ordinal scoring, producing more balanced performance across the two interfaces.

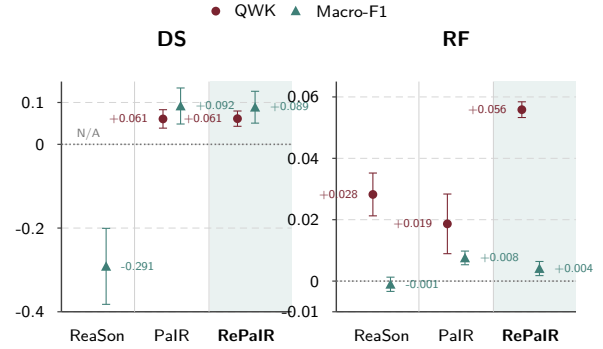


Figure 4: Performance changes relative to standard RS under DS and RF. QWK is averaged across four ordinal tasks and Macro-F1 across three binary tasks; error bars show the standard deviation across seeds. ReaSon–DS QWK is unavailable under strict parsing.

6 Conclusion

This paper examined how supervision objectives and inference protocols jointly affect LLM-based scientific-writing evaluation. Although RS improved teacher-rationale fitting and generated-rationale quality, these gains did not translate into better scoring performance. On ordinal tasks, RF increased severe errors and consequently reduced QWK under both supervision objectives, whereas correcting erroneous rationales recovered most selected harmful cases. PaIR and RePaIR further showed that paired-interface supervision and rationale regeneration can mitigate the tension between scoring performance and interface coverage. These

findings indicate that rationale quality alone is not a reliable proxy for scoring ability, and that supervision objectives should be designed and evaluated together with the intended inference protocol.

7 Limitations.

Our experiments are limited to Qwen3-4B, a fixed LoRA configuration, and seven scientific-writing evaluation tasks. Moreover, rationale quality was assessed by external LLM judges, and the correction experiment covered only selected harmful cases. The findings may therefore not generalize to other models or tasks, nor do they establish that generated rationales faithfully represent internal reasoning. Future work should extend the evaluation to broader models, tasks, and samples, with additional human assessment.

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